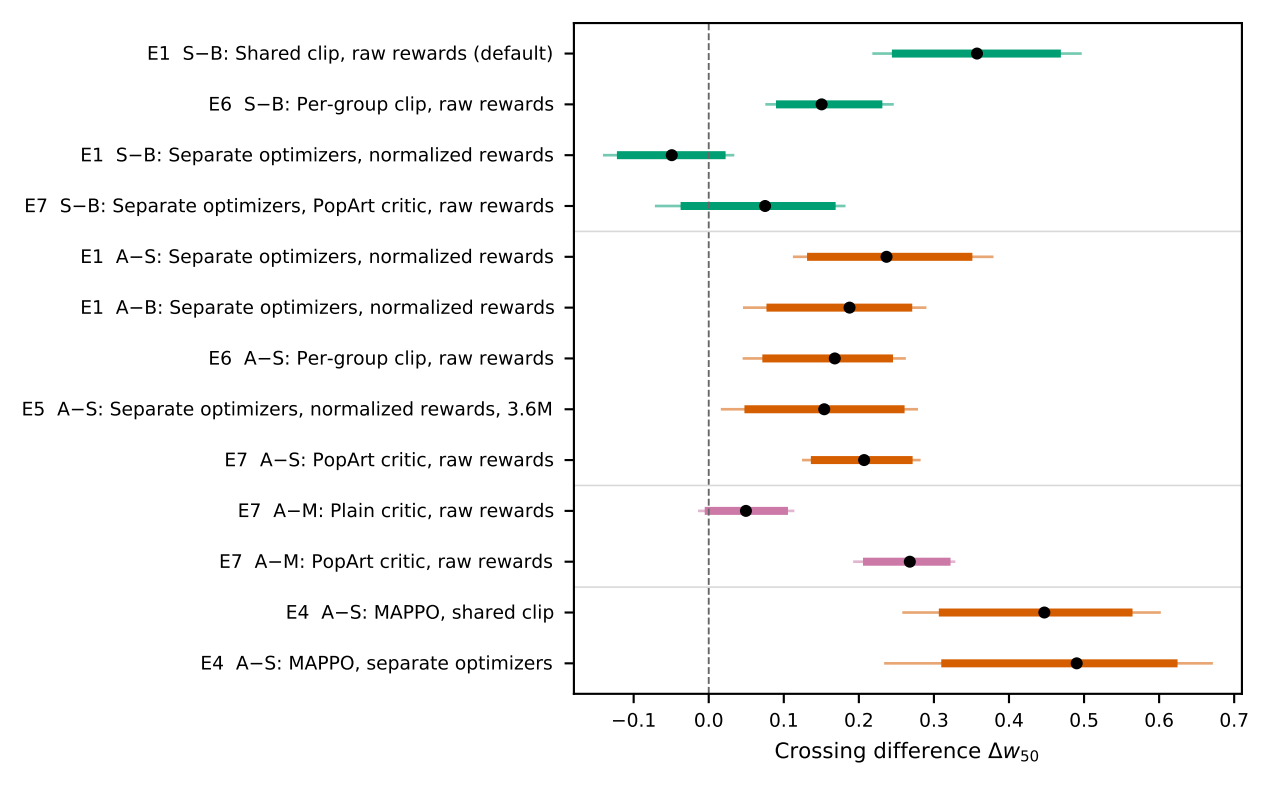
E1 S−B: Shared clip, raw rewards (default)
E6 S−B: Per-group clip, raw rewards
E1 S−B: Separate optimizers, normalized rewards
E7 S−B: Separate optimizers, PopArt critic, raw rewards
E1 A−S: Separate optimizers, normalized rewards
E1 A−B: Separate optimizers, normalized rewards
E6 A−S: Per-group clip, raw rewards
E5 A−S: Separate optimizers, normalized rewards, 3.6M
E7 A−S: PopArt critic, raw rewards
E7 A−M: Plain critic, raw rewards
E7 A−M: PopArt critic, raw rewards
E4 A−S: MAPPO, shared clip
E4 A−S: MAPPO, separate optimizers
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Crossing difference Δw_{50}